

Restocking Calculation Method

Purpose:

- To determine the quantity of items to be replenished at a picking location, taking into account current stock, orders in preparation, and thresholds.

Factors Considered:

1. **Replenishment Unit:** The unit used for replenishment (e.g., pallet, box).
2. **Max Quantity at Picking:** Maximum quantity that can be held at the picking location.
3. **Replenishment Threshold:** Minimum quantity that triggers a replenishment if stock falls below it.
4. **Quantity in Orders in Preparation:** Items already allocated to orders currently in preparation.

Restocking Logic:

1. **Check Orders in Preparation:**
 - If the item exists in orders in preparation and the current picking quantity is less than the maximum at picking, a replenishment is generated.
 - Quantity to replenish = Max at picking – Quantity currently at picking
2. **Check Replenishment Threshold:**
 - If the item is not in orders in preparation:
 - If the current picking quantity < threshold, generate a replenishment.
 - If current picking quantity ≥ threshold, no replenishment is generated.

Calculation Formula:

Need = AvailableQtyPicking – sum(PickingSamplingsNotValidated)
+ sum(REA_NotValidated) – ReplenishmentThreshold

Quantity to Replenish:

$$\text{QtyToReplenish} = \text{PicSamplQty} + \text{MaxPickingQty} - \text{PicAvailableQty} \\ - \text{ReplQty} - \text{QtyTransfToPick} - \text{QtyTransfFromPick}$$

- Adjustments can be made depending on saturation, adding threshold quantity or max picking quantity.
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Example:

- Picking Unit: Pallet (1 pallet = 15 parcels)
- Picking Capacity: 2 pallets
- Threshold: 10% (3 parcels)
- Orders Launched: 45 parcels
- Current Quantity at Picking: 5 parcels

Calculation:

$$\text{Quantity to Replenish} = 45 + 30 - 5 = 70 \text{ parcels}$$

- Optimal quantity to replenish: 70 parcels
- Minimum quantity needed to serve orders: 43 parcels